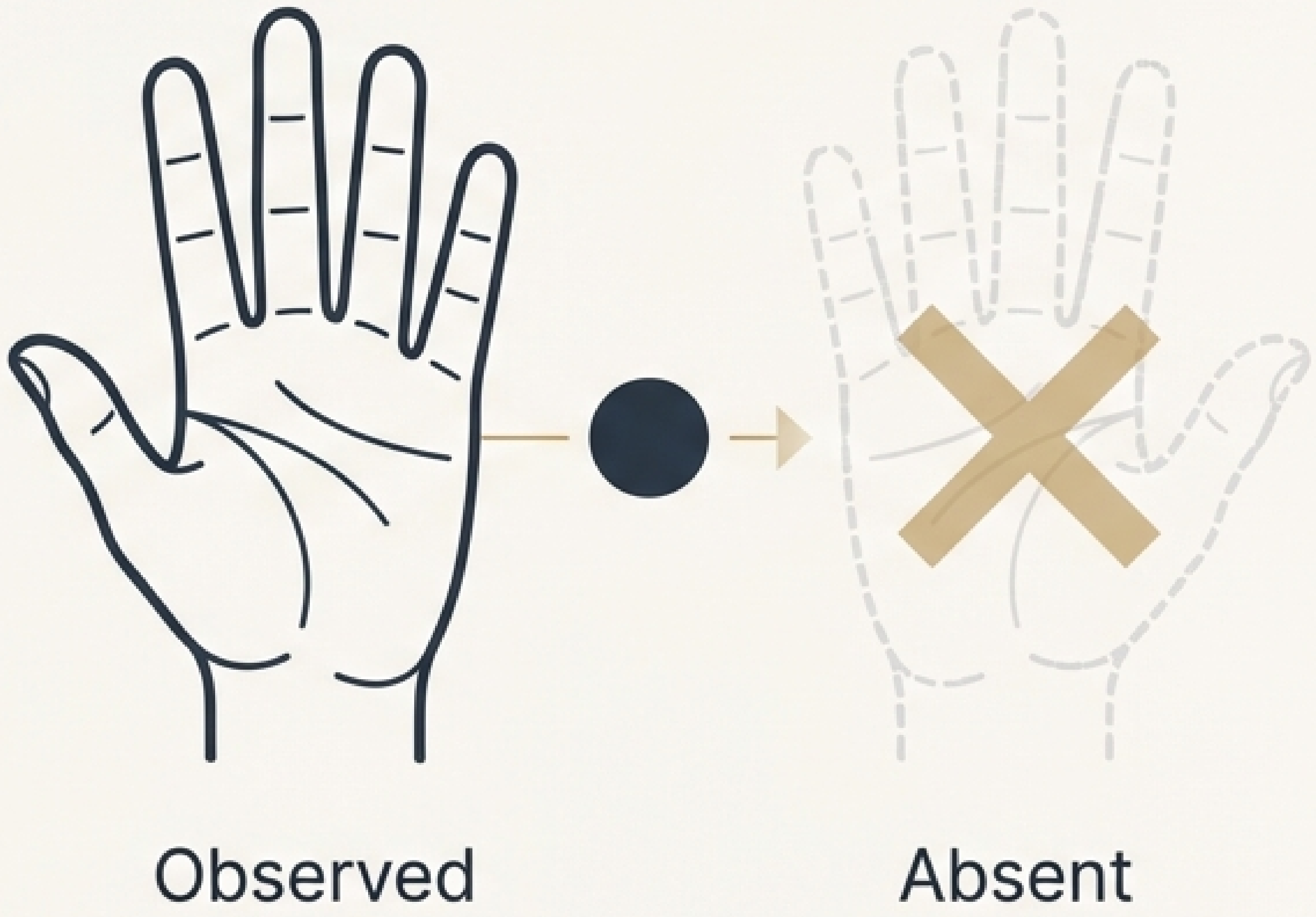
From Rule to Reason: A Geometric Origin for Neutrino Chirality

A First-Principles Derivation in Swirl-String Theory

The Unbroken Left-Handedness of the Neutrino

One of the most profound violations of parity symmetry in nature is an observational fact: in all low-energy interactions, neutrinos appear exclusively left-chiral.

This absolute preference for one “handedness” over the other is a core feature of the weak interaction, yet its fundamental origin remains an open question. Why does nature exhibit this stark asymmetry?



The Standard Model's Answer: An Asymmetry by Construction

The Standard Model successfully accommodates this observation by building the asymmetry directly into its mathematical foundation.

- The electroweak force is governed by the chiral gauge group $SU(2)_L$.
- By definition, only left-chiral fermions (and right-chiral anti-fermions) are assigned a charge under this group.
- Right-chiral neutrinos are simply excluded from the model; they do not interact via the weak force.

> This is an effective description, but it is not an explanation. It is a rule, implemented by construction, rather than a reason that emerges from deeper principles.

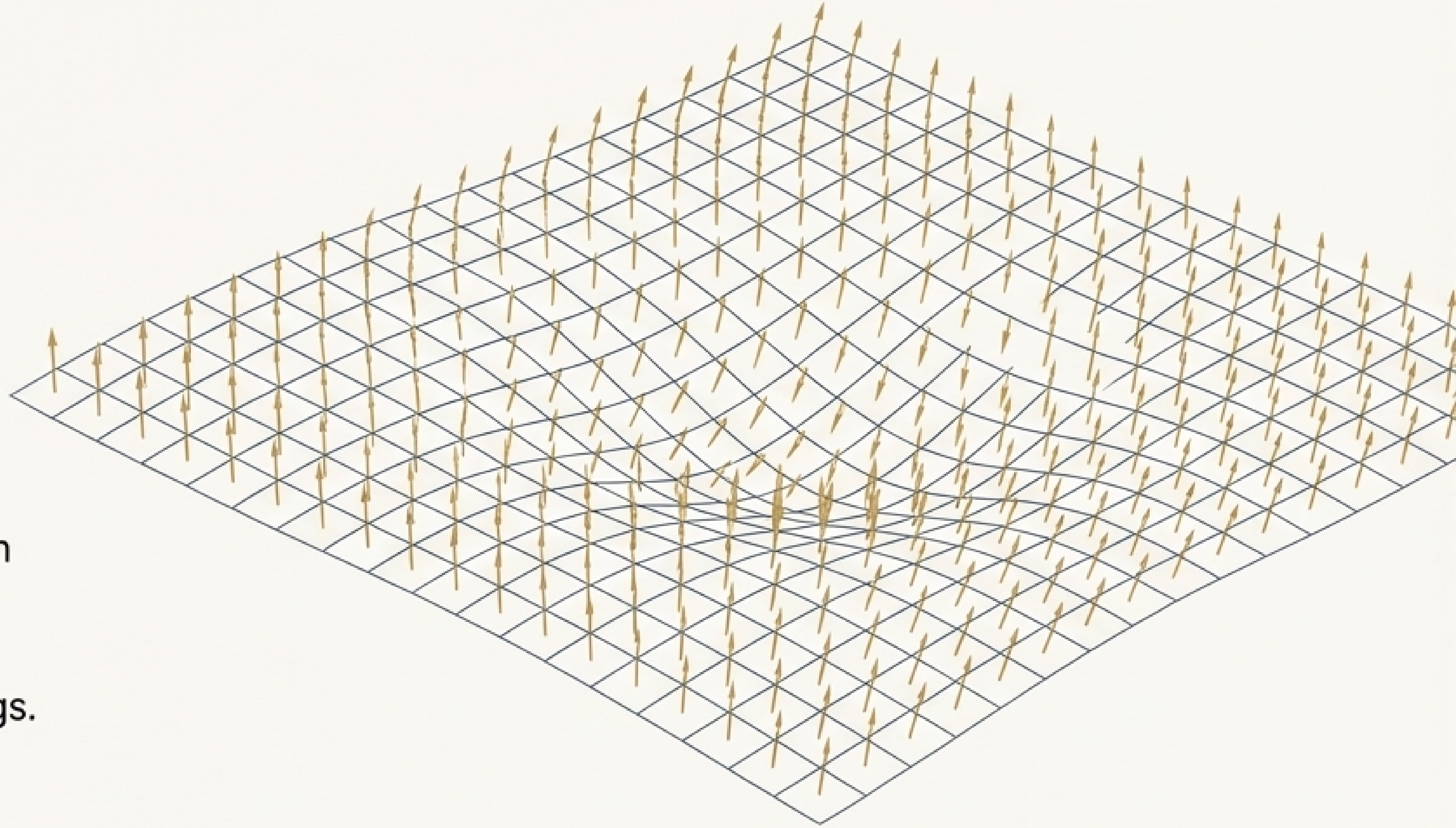
A New Foundation: Spacetime with a Preferred Direction

Swirl-String Theory (SST) proposes a new geometric structure for spacetime: a **global, unit timelike vector field,** u^μ, called the **"swirl-clock field."**

- This field defines a preferred direction at every point, akin to a universal compass needle for spacetime.
- It represents the direction of "vertical swirl" in a background fluid foliation.
- It spontaneously breaks local Lorentz symmetry, allowing for new physical couplings.

$$u^\mu u_\mu = -1$$

A unit vector defining a direction in time



How Spacetime's Compass Filters Chirality

The existence of the u^μ field allows for a new, leading-order interaction with fermions that is consistent with gauge invariance: an axial coupling.

$$L_{\text{int}} = \frac{\lambda}{M^*} u^\mu \underbrace{\bar{\nu}\gamma^\mu\gamma^5\nu}$$

The Geometric Influence — This term imports the direction of the universal compass into the interaction.

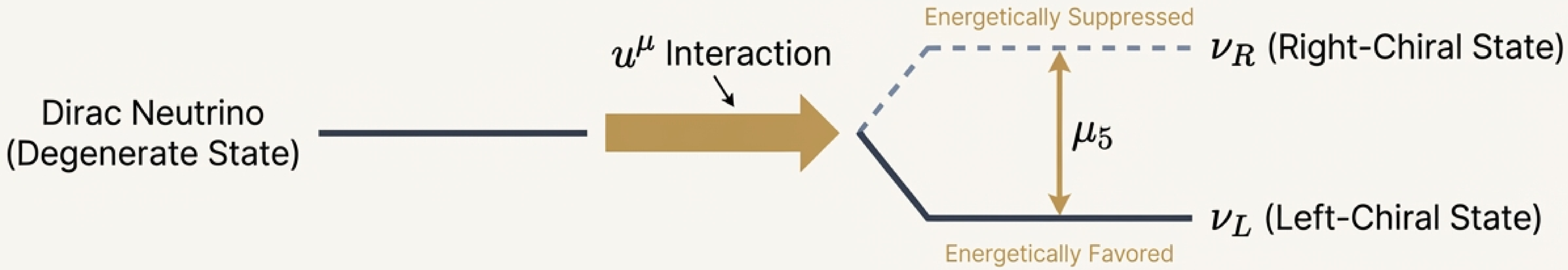
The Energy Scale of the Interaction — A crucial parameter that determines the strength of strength of the effect. At this point, it is an unknown scale we must derive from the theory.

The Chirality Filter — This is the axial-vector current, which naturally separates left- and right-chiral components.

The Result: A Dynamic and Energetic Divide

This axial coupling acts like a "chiral chemical potential," altering the energy of the two neutrino states. In the rest frame of the foliation ($u^\mu \approx (1,0,0,0)$), the interaction term simplifies, causing an energy shift.

$$E_{L,R}(p) = |p| \mp \mu_5, \text{where } \mu_5 = \frac{\lambda}{M^*}$$



If the energy gap μ_5 is sufficiently large, the right-chiral state is effectively "integrated out" of the low-energy theory. Neutrinos naturally appear left-handed.

Anchoring the Theory in the Electron

The mechanism's validity depends on the unknown scale M^*. Can it be derived from first principles within SST?

SST defines a fundamental 'swirl-force scale" derived from the harmonic motion of the electron. It connects two of the electron's most fundamental properties:

- Its quantum clock: The Compton frequency, $\omega_C = \frac{m_e c^2}{\hbar}$.
- Its classical scale: The classical electron radius, $r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2}$.

e^-

Compton Frequency (ω_C)

$$\omega_C = \frac{m_e c^2}{\hbar}$$

Classical Radius (r_e)

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2}$$

The Swirl-Force Scale

$$F_{swirl}^{\max} = m_e \left(\frac{\omega_C}{\alpha}\right)^2 r_e$$

The Result: The Chiral Scale is Half an Electron's Mass

Connecting Force to Energy: Within SST, this force scale defines a characteristic energy, E_{swirl}. Calculation from first principles yields a stunningly simple result:

$$\boxed{E_{swirl} \approx \frac{1}{2} m_e c^2}$$

$$E_{swirl} \approx 2.555 \times 10^5 \text{ eV}$$

The energy scale of the axial coupling, M^*, is identified with this swirl energy scale.

$$M^* = \frac{E_{swirl}}{c^2} = \frac{1}{2} m_e$$

The force that suppresses right-handed neutrinos is not an arbitrary new scale. It is determined by the fundamental physics of the electron.

A Universal Selection Rule for Matter

The Connection to Hydrogen: In SST, particles carry "swirl-charge." Hydrogen's swirl-charge is determined by its constituents (3 swirl strings: proton components + electron), giving it a net positive alignment.

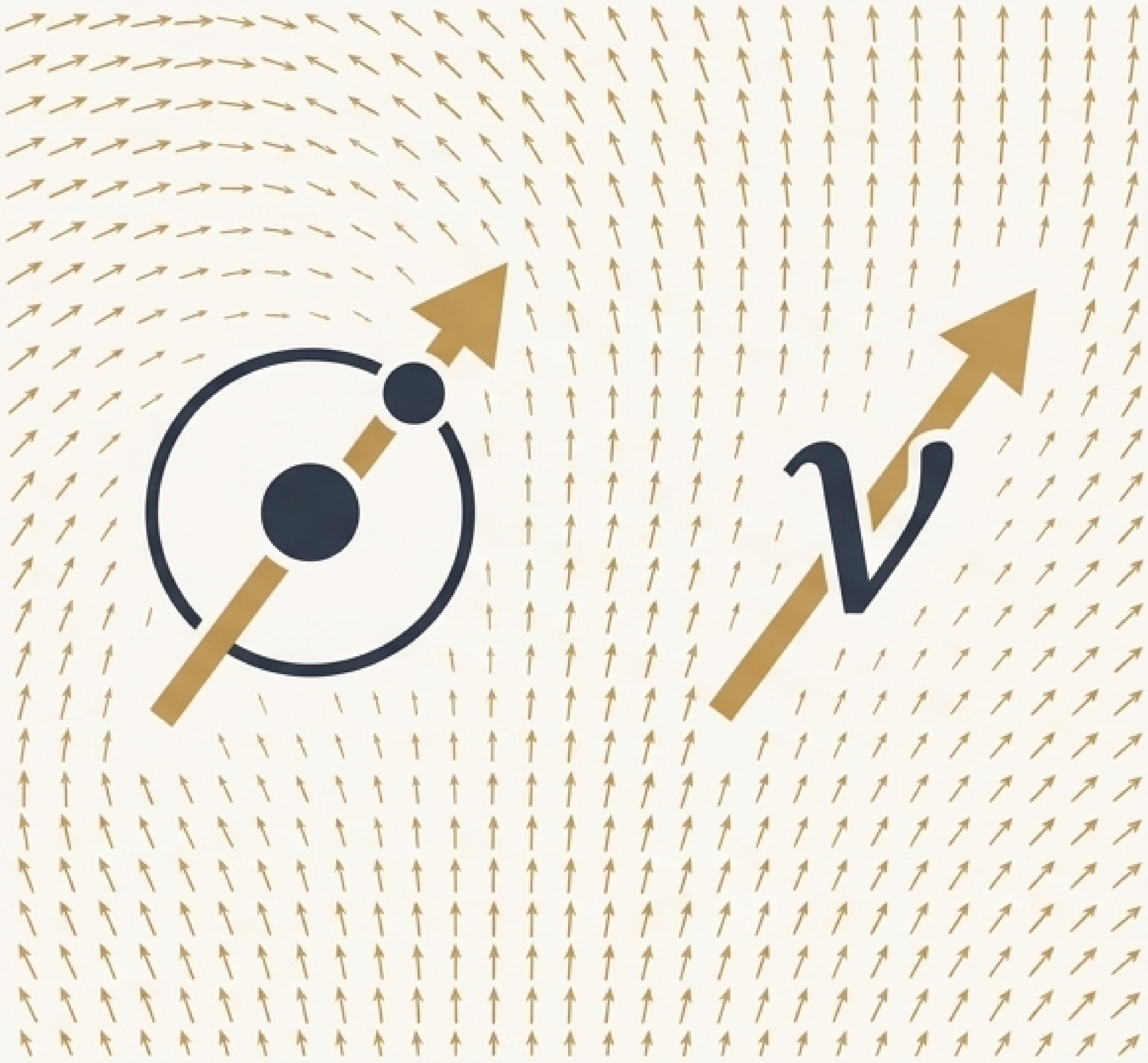
$$Q_{swirl}(H) = +3$$

The Neutrino's Place: The neutrino is the minimal particle that obeys this same alignment principle.

$$Q_{swirl}(\nu) = +1$$

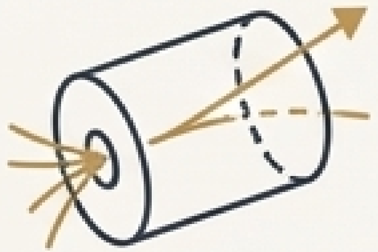
The Unifying Principle: Both Hydrogen and the left-handed neutrino align with the same universal swirl-clock direction. The electroweak chiral asymmetry is a direct consequence of this universal rule:

Preferred Chirality $\Leftrightarrow$ Alignment with u^μ.



Four Falsifiable Signatures to Test Swirl-String Theory

SST's geometric mechanism leads to specific, observable predictions that distinguish it from the Standard Model.

1. 

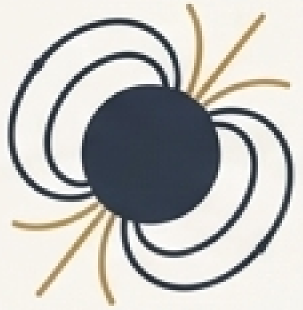
Right-Handed Neutrino Suppression

Searches for sterile neutrinos should find an effective high mass gap, set by the scale $\mu_5 \propto \lambda/m_e$.

2. 

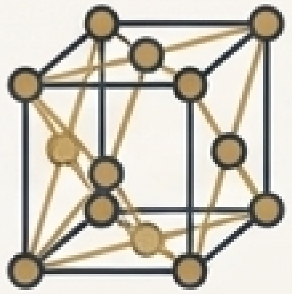
Directional Modulation

A cosmological variation in the u^μ field would create directional asymmetries in the flux of ultra-high-energy cosmic neutrinos.

3. 

Chiral Amplification in Gravitational Wells

The effective chiral potential μ_5 should increase near objects with high swirl density, such as neutron stars, enhancing parity-violating effects.

4. 

Hydrogen-Dependent Chirality Lock-In

Materials with coherent Hydrogen swirl-charge (e.g., metallic hydrides) may exhibit enhanced parity-violating optical activity.

Conclusion: Chirality as a Consequence of Geometry

- The left-handedness of the neutrino is not an imposed gauge asymmetry, as in the Standard Model.
- It is a dynamical consequence of a universal axial coupling to a geometric field, u^{μ}, that defines a preferred direction in spacetime.
- The strength of this coupling is not a free parameter; it is anchored directly to the harmonic motion and scale of the electron.

SST provides a parsimonious, first-principles explanation, unifying the chiral asymmetry of the electroweak sector with the fundamental structure of matter and spacetime.